SEMINAR 1

Temă: 1) Se aruncă 6 zaruri. Notăm cu A evenimentul fericit: suma zarurilor $\in [15, 20]$. Care este probabilitatea lui A?

2) Câte tipuri de zaruri se pot fabrica (toate fețele trebuie să fie echiprobabile). Se poate face un zar cu 5 fețe? (cu 5 e tetraedru regulat)

1) Se consideră ecuația de gradul 2 $x^2+bx+c=0$, unde $b, c \in [0,4]$. Notăm cu A evenimentul ca ambele rădăcini să fie reale.

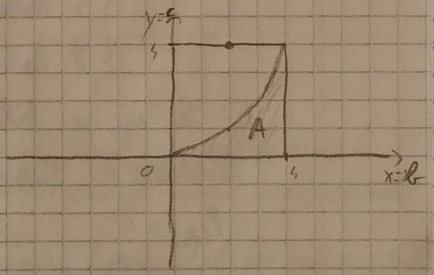
B: evenimentul ca $x_1 = x_2$

C: evenimentul ca rădăcinile să fie complexe

$P(A) = ?$; $P(B) = ?$; $P(C) = ?$

Rez: $\Delta = b^2 - 4c$.

Probabilități geometrice



$A: b^2 - 4c \geq 0 \Rightarrow c \leq \frac{b^2}{4}$

$P(A) = \frac{\text{Aria favorabilă}}{\text{Aria tot}} = \frac{A_f}{16}$

$A_f = \int_0^4 \frac{x^2}{4}\,dx = \frac{1}{4}\cdot\frac{x^3}{3}\Big|_0^4 = \frac{1}{4}\cdot\frac{64}{3} = \frac{16}{3}$

$P(A) = \frac{16}{3}\cdot\frac{1}{16} = \frac{1}{3}$

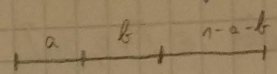
$B: b^2 - 4c = 0$; $P(B) = 0$ (graficul e o dreaptă); B se numește miracol (eveniment care se poate produce, dar cu probabilitate 0)

$C:$ $P(C) = 1$

C se numește eveniment aproape sigur (e sigur de fapt)

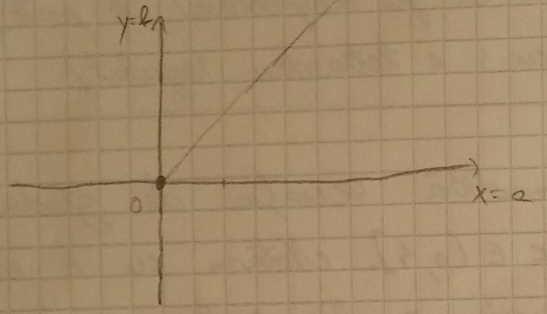
$D: x_1 \neq x_2$; $P(D) = 1$; D aproape sigur

Temă: 3) Un chinez vrea să facă triunghiuri și angajează mulți chinezi și are la dispoziție bare de 1 m. Barele sunt tăiate în 2 puncte aleatoare. Care e probabilitatea ca din cele 3 bucăți să se formeze un triunghi?



$0 < a < b < 1-a-b \Rightarrow \begin{cases} b < 1-2a \\ b < \frac{1}{2} - \frac{1}{2}a \end{cases}$

$1-a-b < a+b \Rightarrow a+b > \frac{1}{2} \Rightarrow b > \frac{1}{2} - a$



Schema lui Bernoulli fără revenire

Ex.: U e o urnă

U: { 5 bile de tip a
4 bile de tip b
8 bile de tip c

Pascal - Fermat
$P(A) = \frac{\text{nr. cazuri favorabile}}{\text{nr. cazuri posibile}}$

E: Se ia o bilă și se pune alături. Se face E de 3 ori.

A: Evenimentul de a avea 3 bile două de 2 tipuri

$P(A) = ?$

nr. tot. de cazuri: $C_{17}^3 = \frac{17 \cdot 16 \cdot 15}{6} = 17 \cdot 8 \cdot 5 = 680$

nr. cazuri favorabile: $C_5^2 \cdot C_4^1 + C_5^2 \cdot C_8^1 + C_4^2 (C_5^1 + C_8^1) + C_8^2 (C_5^1 + C_4^1) = 10 \cdot 12 + 6 \cdot 13 + 28 \cdot 9 = 120 + 78 + 252 = 450$

$450 \mid \frac{68}{6}$

$P(A) = \frac{450}{680} = \frac{45}{68} = 0,6$

Temă: 4) Schema lui Bernoulli cu revenire (se pun înapoi bilele extrase)

nr. tot. de cazuri:

nr. cazuri favorabile:

Def.: Un număr x real se numește normal dacă în scrierea sa zecimală, orice zecimală apare cu o probabilitate de $\frac{1}{10}$.

3.03.2015

SEMINAR 2

$$(x_0 + x_1^2 + \dots + x_6^5)^6 =$$
$$= x^6(1 + x + x^2 + x^3 + x^4)^6$$

1) Se aruncă 10 zaruri. Se notează cu A: să apară toate fețele. $P(A) = ?$

$$P(A) = \frac{A_{10}^6 \cdot 6^4}{6^{10}} = \frac{A_{10}^6}{6^6}$$

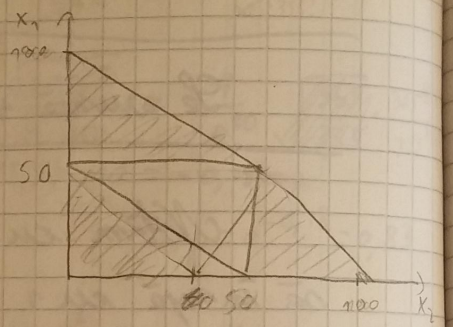
$$x^6 \cdot \left(\frac{1-x^5}{1-x}\right)^6$$

2) O firmă cu 3 acționari. Probabilitatea ca firma să aibă acționariat majoritar

$$\begin{cases} x_1, x_2, x_3 \geq 0 \\ x_1 + x_2 + x_3 = 100 \end{cases}$$

A: $x_1 \geq 50$ sau $x_2 \geq 50$ sau $x_3 \geq 50$

$$\begin{cases} x_1 \geq 0 \\ x_2 \geq 0 \\ 100 - x_1 - x_2 \geq 0 \end{cases}$$



$$A: \begin{cases} x_1 \geq 50 \\ \text{sau} \\ x_2 \geq 50 \\ \text{sau} \\ 100 - x_1 - x_2 \geq 50 \Rightarrow x_1 + x_2 \leq 50 \end{cases}$$

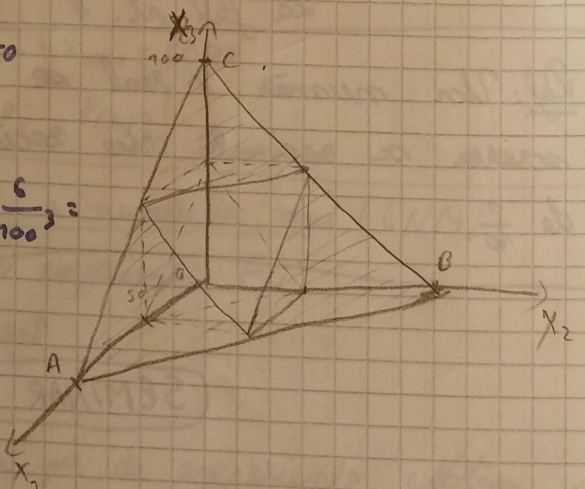
$$P_A = \frac{A_R}{A_t} = 75\%$$

---

3) Firmă cu 4 acționari

$$\begin{cases} x_1 + x_2 + x_3 + x_4 = 100 \\ x_1, x_2, x_3, x_4 \geq 0 \end{cases} \Leftrightarrow \begin{cases} x_1, x_2, x_3 \geq 0 \\ 100 - x_1 - x_2 - x_3 \geq 0 \Rightarrow x_1 + x_2 + x_3 \leq 100 \end{cases}$$

$$A: \begin{cases} x_1 \geq 50 \\ \text{sau} \\ x_2 \geq 50 \\ \text{sau} \\ x_3 \geq 50 \\ \text{sau} \\ x_1 + x_2 + x_3 \leq 50 \end{cases}$$



$$P_{(A)} = \frac{V_R}{V_{tot}} = \frac{4 \cdot 50^3}{6} \cdot \frac{6}{100^3} = \frac{4}{8} = \frac{1}{2}$$

Temă: pt. 5 acționari

---

4) Se aruncă cu 4 zaruri. A: suma zarurilor $\in [12, 22]$

Legea numerelor mari:

1: 1, 2, 3, 6 ; $\Sigma = 12$; 1 $\longrightarrow$ 1

2: 6, 4, 4, 5 ; $\Sigma = 19$; 1 $\longrightarrow$ 0,5

3: 1, 2, 4, 5 ; $\Sigma = 12$; 2 $\longrightarrow$ 0,66

4: 2, 4, 5, 6 ; $\Sigma = 17$; 3 $\longrightarrow$ 0,75

5: 2, 4, 5, 6 ; $\Sigma = 17$; 4 $\longrightarrow$ 0,8

6: 3, 3, 4, 1 ; $\Sigma = 11$; 4 $\longrightarrow$ 0,66

7: 6, 2, 3, 2 ; $\Sigma = 13$ 5 $\longrightarrow$ 0,71

8: 6, 6, 3, 3 ; $\Sigma = 18$; 5 $\longrightarrow$ 0,62

9: 5, 5, 4, 2 ; $\Sigma = 16$; 6 $\longrightarrow$ 0,66

10: 4, 4, 2, 2 ; $\Sigma = 12$; 7 $\longrightarrow$ 0,7

Schema lui Poisson (generalizare Bernoulli cu revenire)

Considerăm 3 urne.

U_1: 5a, 3b, 2c ; se ia o bilă

U_2: 4a, 2b, 1c, 3d ; se ia o bilă

U_3: 3a, 3b, 2c ; se ia o bilă

A: doar 2 tipuri de bile ; $P(A) = ?$

$P(x,y,z,u) = \left(\frac{1}{2}x + \frac{3}{10}y + \frac{1}{5}z\right)\left(\frac{2}{5}x + \frac{1}{5}y + \frac{1}{10}z + \frac{3}{10}u\right)\left(\frac{3}{8}x + \frac{3}{8}y + \frac{1}{4}z\right)$

$P(A)$ = suma coeficienților monoamelor de 2 variabile.

$$P(A) = \frac{1}{2}\cdot\frac{2}{5}\cdot\frac{3}{8} + \frac{1}{2}\cdot\frac{2}{5}\cdot\frac{1}{4} + \frac{1}{2}\cdot\frac{3}{8}\cdot\frac{1}{5} + \frac{1}{2}\cdot\frac{3}{8}\cdot\frac{1}{10} + \frac{1}{2}\cdot\frac{3}{8}\cdot\frac{3}{10} + \frac{2}{5}\cdot\frac{3}{8}\left(\frac{3}{10}+\frac{1}{5}\right) + \frac{3}{10}\cdot\frac{1}{5}\left(\frac{3}{8}+\frac{1}{4}\right) + \frac{3}{10}\cdot\frac{3}{8}\left(\frac{2}{5}+\frac{1}{10}+\frac{3}{10}\right) + \frac{1}{5}\cdot\frac{3}{8}\left(\frac{1}{2}+\frac{1}{5}\right) + \frac{1}{5}\cdot\frac{1}{10}\left(\frac{3}{8}+\frac{3}{8}\right) + \frac{1}{5}\cdot\frac{1}{4}\left(\frac{2}{5}+\frac{1}{5}+\frac{3}{10}\right) + \frac{1}{10}\cdot\frac{1}{4}\left(\frac{1}{2}+\frac{3}{10}\right) = \frac{758}{800} = 0,57$$

$P(A_{p_n}) = \frac{1}{p_n}$ și $P(\bar{A}_{p_n}) = 1 - \frac{1}{p_n}$

X = ev. de a fi nr. prim.

$X = \bar{A}_2 \cap \bar{A}_3 \cap \dots \cap \bar{A}_{p_n}$

$P(X) = P(\bar{A}_2) \cdot \dots \cdot P(\bar{A}_{p_n}) \Rightarrow \frac{1}{P(X)} = \frac{1}{1-\frac{1}{2}} \cdot \frac{1}{1-\frac{1}{3}} \cdot \dots \cdot \frac{1}{1-\frac{1}{n}} \Big\} \Rightarrow$

$\frac{1}{1-x} = \sum_{n=1}^{\infty} x^n$

$\Rightarrow \frac{1}{P(X)} \to \infty \Rightarrow P(X) \to 0$

10.03.2015

SEMINAR 3

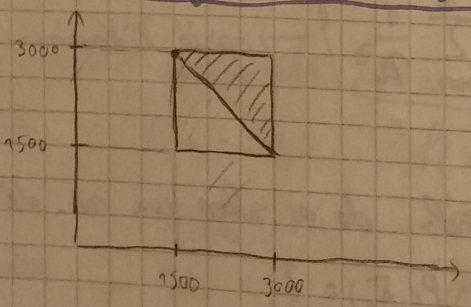
1) Să se arate că există secvențe consecutive oricât de lungi în care nu se află un număr prim. (10 nr.)

2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61

2) Doi tineri câștigă între 1500 RON și 3000 RON. Care e probabilitatea să depășească suma salariului 4500.

3) Se numesc nr. prime gemene 2 nr. prime $|m-n|=2$. Să se arate că există o infinitate de astfel de nr.

2)



$P = \frac{A_R}{A_{tot}} = \frac{1}{2}$

4) Fie A, B, C 3 evenimente independente în câmpul de probabilitate (Ω, K, p). $P(A)=0,8$. $P(B)=0,3$. $P(C)=0,8$.

$P(A \cup B \cup C) = ?$; $P(A \cup B \cup C) = P(A)+P(B)+P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A$

$P(A \cap B \cap C) = ?$

$P(A \cup B \mid C) = ?$

$P(A \cup B \mid C) = \frac{P((A \cup B) \cdot P(C)}{P(C)} \quad \frac{P((A \cup B) \cap C)}{P(C)}$

$P((A \cup B) \cap C) = P((A \cap C) \cup (B \cap C)) = P(A \cap C) + P(B \cap C) - P(A \cap B \cap C) =$

$= P(A) \cdot P(C) + P(B) \cdot P(C) - P(A) \cdot P(B) \cdot P(C) = (P(A) + P(B) - P(A) \cdot P(B)) \cdot P(C)$ $\Big\} =$

$P(A \cup B) \cdot P(C) = (P(A) + P(B) - P(A \cap B)) \cdot P(C) = (P(A) + P(B) - P(A \cap B)) \cdot P(C)$

$\Rightarrow P((A \cup B) \cap C) = P(A \cup B) \cdot P(C)$

---

1) $(n+1)! + 2 + \dots (n+1)! + (n+1) \dots$

12.03.2015

CURS 3. Variabile aleatoare

Fie (Ω, K, p) triplet al lui Kolmogorov. Fie $(A_1, A_2 \dots A_n)$ sistem complet de evenimente.

O funcție $f: \Omega \to \mathbb{R}$ a.î. $f(\omega) = \begin{cases} \alpha_1 ; \omega \in A_1 \\ \alpha_2 ; \omega \in A_2 \\ \vdots \\ \alpha_n ; \omega \in A_n \end{cases}$

se numește variabilă aleatoare simplă cu evenimentele de nivel $A_1, A_2 \dots A_n$.

$$\int_{-\infty}^{\beta} P(x)dx - \int_{-\infty}^{\alpha} P(x)dx = \int_{\alpha}^{\beta} P(x)dx$$

17.03.2015

SEMINAR 3

1) Se aruncă cu 2 zaruri. Să se det. var aleatoare care monitorizează nr. de apariții a cifrei 6. Să se det F_f, M_f, X_f, D^2_f, σ_f.

$(\Omega, \mathcal{K}, P)$ → camp de probabilitate

$$f(\omega) = \begin{cases} 0 & \omega \in \{\dots\}; A_1 \\ 1 & \omega \in \{(1,6),(2,6)\dots(5,6);(6,1);(6,2)\dots;(6,5)\}; A_2 \\ 2 & \omega \in \{(6,6)\}; A_2 \end{cases}$$

$f: \Omega \to \mathbb{R}$; $f(\omega) = \begin{cases} 0; \omega \in A_1 \\ 1; \omega \in A_2 \\ 2; \omega \in A_3 \end{cases} \Leftrightarrow f = 0 \cdot \chi_{A_1} + 1 \cdot \chi_{A_2} + 2 \cdot \chi_{A_3}$

$F: \mathbb{R} \to \mathbb{R}$
$F(x) = P\{\omega \in \Omega \mid f(\omega) < x\}$

$$F(x) = \begin{cases} 0; & x \le 0 \\ \frac{25}{36}; & x \in (0,1] \\ \frac{35}{36}; & x \in (1,2] \\ 1; & x > 2 \end{cases}$$

prb. (lege)
are lege
ulati

$n = 70$
$\sigma = 4$

1 PEARSON 2

$$X_g = \begin{pmatrix} 0 & 1 & 2 \\ \frac{25}{36} & \frac{10}{36} & \frac{1}{36} \end{pmatrix} = \begin{pmatrix} x_1 & x_2 & x_3 \\ p_1 & p_2 & p_3 \end{pmatrix}$$

$$M_g = \sum_{i=1}^{3} x_i p_i = \frac{10}{36} + \frac{2}{36} = \frac{1}{3}$$

$$D_g^2 = \sum_{i=1}^{3} x_i^2 p_i - M_g^2 = \frac{10}{36} + \frac{4}{36} - \frac{1}{9} = \frac{14}{36} - \frac{4}{36} = \frac{10}{36}$$

$$\sigma_g = \sqrt{D_g^2} = \frac{\sqrt{10}}{6}$$

---

2) Problema 1 cu 3 zaruri.

$$f(\omega) = \begin{cases} 0; & \omega \in A_1 \rightarrow \text{even. ca niciun zar} \\ & \text{nu e alb} \\ 1; & \omega \in A_2 \; ; \; \omega_1 = 6; \omega_2 \neq 6, \omega_3 \neq 6 \cdot - \\ 2; & \omega \in A_3 \; ; \; \omega_1 = \omega_2 = 6 \; ; \; \omega_3 \neq 6 \ldots \\ 3; & \omega \in A_4 \; ; \; \omega_1 = \omega_2 = \omega_3 = 6 \end{cases}$$

$$X_g = \begin{pmatrix} 0 & 1 & 2 & 3 \\ \frac{125}{216} & \frac{75}{216} & \frac{15}{216} & \frac{1}{216} \end{pmatrix}$$

$$M = \frac{75}{216} + \frac{30}{216} + \frac{3}{216} = \frac{108}{216} = \frac{1}{2}$$

$$D_f^2 = \frac{75}{216} + \frac{60}{216} + \frac{9}{216} - \frac{1}{4} = \frac{144}{216} - \frac{1}{4} = \frac{6}{9} - \frac{1}{4} = \frac{2}{3} - \frac{1}{4} = \frac{5}{12}$$

$$\sigma_f = \sqrt{\frac{5}{12}}$$

$$F: \mathbb{R} \rightarrow \mathbb{R}$$

$$F(x) = \begin{cases} 0; & x < 0 \\ \frac{125}{216}; & x \in [0,1) \\ \frac{200}{216}; & x \in [1,2) \\ \frac{215}{216}; & x \in [2,3) \\ 1; & x \geq 3 \end{cases}$$

---

3) Se aruncă 5 zaruri. Se not. cu S suma zarurilor. Să se det. prob. ca suma zarurilor să fie între [15, 20].

Notăm cu f_i v.a. care monitorizează zarul i.

$$X_{f_i} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \end{pmatrix}$$

$$X_{f_1+f_2} = \begin{pmatrix} 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 \\ \frac{1}{36} & \frac{2}{36} & \frac{3}{36} & \frac{4}{36} & \frac{5}{36} & \frac{6}{36} & \frac{5}{36} & \frac{4}{36} & \frac{3}{36} & \frac{2}{36} & \frac{1}{36} \end{pmatrix}$$

~~$X_{f_1+f_2+f_3+f_4} = \begin{pmatrix} 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 \end{pmatrix}$~~

$$X_{f_1+f_2+f_3+f_4} = \begin{pmatrix} \dots & 9 & 10 & 11 & 12 & 13 & 14 & 15 & 16 & 17 & 18 & 19 & \dots \\ & \frac{6}{1296} & & & & & & & & & & & \end{pmatrix}$$

CURS 9

Variabile aleatoare continue

(Ω, K, P) — câmp de probabilitate

24.03.2015

SEMINAR 5

1) Fie f, g - 2 v.a indep. simple.

$X_f = \begin{pmatrix} -1 & 0 & 1 \\ 0,2 & 0,3 & 0,5 \end{pmatrix}$; $X_g = \begin{pmatrix} -1 & 1 \\ 0,4 & 0,6 \end{pmatrix}$

$F(x) = P(f(\omega) \leq x)$

$F(y) = P(g(\omega) \leq y)$

a) Să se det. F_f, F_g

b) Să se det. M_f, $D^2 f$, σ_f, M_g, $D^2 g$, σ_g

c) Det. M_{f+g}, D^2_{f+g}, σ_{f+g}

a) $F_f(x) = \begin{cases} 0; & x \in (-\infty; -1) \\ 0,2; & x \in [-1; 0) \\ 0,5; & x \in [0, 1) \\ 1; & x \in [1, \infty) \end{cases}$; $F_g(x) = \begin{cases} 0; & x \in (-\infty; -1) \\ 0,4; & x \in [-1; 1) \\ 1; & x \in [1; \infty) \end{cases}$

b) $M_f = -1 \cdot 0,2 + 0 \cdot 0,3 + 1 \cdot 0,5 = 0,3$

$D^2_f = (-1)^2 \cdot 0,2 + 0^2 \cdot 0,3 + 1^2 \cdot 0,5 - M_f^2 = 0,7 - 0,09 = 0,61$

$\sigma_f = \sqrt{D^2 f} = \sqrt{0,61} = 0,78$

$M_g = -0,4 + 0,6 = 0,2$

$D^2_g = 0,4 + 0,6 - 0,04 = 0,96$

$\sigma_g = \sqrt{D^2_g} = 0,98$

c) $X_{f+g} = \begin{pmatrix} -2 & -1 & 0 & 1 & 2 \\ 0,08 & 0,12 & 0,12+0,2=0,32 & 0,18 & 0,3 \end{pmatrix}$

$M_{f+g} = -0,16 - 0,12 + 0,18 + 0,6 = 0,5$

$D^2_{f+g} = 4 \cdot 0,08 + 1 \cdot 0,12 + 0,18 + 4 \cdot 0,3 - 0,25 =$

$= 0,32 + 0,12 + 0,18 + 1,2 - 0,25 = 1,82 - 0,25 =$

$D^2_{f+g} = 1,57$

$\sigma_{f+g} = \sqrt{D^2_{f+g}} = 1,25$

2) Le dau 2 v.a. continue și indep.

$f(w)$ uniform distribuită în $[0,2]$

$g(w)$ exponențial distribuită cu $\lambda = 3$ la dreapta

a) $m(f)$, $D^2(f)$, $\sigma(f)$

b) $m(g)$, $D^2(g)$, $\sigma(g)$

c) $m(f+g)$, $D^2(f+g)$, $\sigma(f+g)$

d) se det. pb. ca $f+g \le 2$

a) ~~$f(w) = \begin{cases} \frac{1}{2}; & x \in [0,2] \\ 0; & x \notin [0,2] \end{cases}$~~

$$p_f(x) = \begin{cases} \frac{1}{2}; & x \in [0,2] \\ 0; & x \notin [0,2] \end{cases}$$

$$m_f = \int_{-\infty}^{\infty} x\, p_f(x)\, dx = \frac{1}{2}\int_0^2 x\, dx = \frac{1}{2}\cdot \frac{x^2}{2}\Big|_0^2 = 1$$

$$D^2(f) = \int_{-\infty}^{\infty} x^2 p_f(x)\, dx - m_f^2 = \frac{1}{2}\int_0^2 x^2 dx = \frac{1}{2}\cdot \frac{x^3}{3}\Big|_0^2 - 1 = \frac{4}{3} - 1 = \frac{1}{3}$$

$$\sigma(f) = \sqrt{D^2(f)} = \frac{1}{\sqrt{3}}$$

b) $p_g(w) = \begin{cases} 0; & x \le 0 \\ 3e^{-3x}; & x \ge 0 \end{cases}$

$$m_g = \int_{-\infty}^{\infty} x\, p_g(x)\, dx = 3\int_0^{\infty} x e^{-3x} dx = 3\left(-\frac{1}{3}x e^{-3x}\Big|_0^{\infty} - \frac{1}{9}e^{-3x}\Big|_0^{\infty}\right) = \frac{1}{3}$$

$-\frac{2}{3}xe^{-3x}$ $\quad -\frac{2}{9}e^{-3x}$

$$D^2(g) = \int_{-\infty}^{\infty} x^2 p_g(x)\, dx - m_g^2 = 3\int_0^{\infty} x^2 e^{-3x} dx - \frac{1}{9} = 3\left(-\frac{1}{3}x^2 e^{-3x}\Big|_0^{\infty} + \frac{2}{3}\cdot\left(-\frac{1}{3}\right)xe^{-3x}\Big|_0^{\infty} + \left(+\frac{2}{3}\right)\cdot\left(-\frac{1}{3}\right)e^{-3x}\Big|_0^{\infty}\right) - \frac{1}{9} = \frac{2}{9} - \frac{1}{9} = \frac{1}{9}$$

$$\sigma(g) = \sqrt{D^2(g)} = \frac{1}{3}$$

c) $m(f+g) = m(f) + m(g) = \frac{4}{3}$

$D^2(f+g) = D^2(f) + D^2(g) = \frac{4}{9}$

$\sigma(f+g) = \sqrt{D^2(f+g)} = \frac{2}{3}$

$\Rightarrow \Phi\left(\frac{0,06\sqrt{n}}{\sqrt{5}}\right) - \Phi\left(\frac{-0,06\sqrt{n}}{\sqrt{5}}\right) \geq 0,8 \Rightarrow$

$\Rightarrow \Phi(0,027\sqrt{n}) \geq 0,9 \Rightarrow$

$\Rightarrow 0,027\sqrt{n} > 1,29 \Rightarrow \sqrt{n} > 47,77 \Rightarrow n \geq 2283$

SEMINAR 6

31.03.2015

Se aruncă cu 3 zaruri: A → ev. ca suma zarurilor ≥ 10.

a) $P(A) = ?$

b) v.a. care numără acest ev.

c) Se face exp. de 500 ori. Care e prob. ca acest ev. să se producă între 300 și 500 ori.

d) $n = ?$ a.î. $\nu_n = p$ cu eroare 0,05 și gf. să fie adev. cu o prob. de 0,9

a)

$X_{f_1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \end{pmatrix}$ f_1, f_2, f_3 indep. și din aceeași lume

$X_{f_1+f_2} = \begin{pmatrix} 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & \dots \\ \frac{1}{36} & \frac{2}{36} & \frac{3}{36} & \frac{4}{36} & \frac{5}{36} & \frac{6}{36} & \frac{5}{36} & \frac{4}{36} & \end{pmatrix}$

$X_{f_1+f_2+f_3} = \begin{pmatrix} 3 & 4 & 5 & 6 & 7 & 8 & 9 & \dots \\ \frac{1}{6^3} & \frac{3}{6^3} & \frac{6}{6^3} & \frac{10}{6^3} & \frac{15}{6^3} & \frac{21}{6^3} & \frac{25}{6^3} & \end{pmatrix}$

$P(\bar{A}) = \frac{1+3+6+10+15+21+25}{6^3} = \frac{81}{6^3} = \frac{3}{8}$

$P(A) = 1 - P(\bar{A}) = \frac{5}{8} = 0,625 \simeq 0,63$

b) $f(\omega) = \begin{cases} 0, & A \text{ nu se petrece} \\ 1, & A \text{ se petrece} \end{cases}$; $X_f = \begin{pmatrix} 0 & 1 \\ 0,37 & 0,63 \end{pmatrix}$

$m_f = 0,63$

$D_f^2 = 0,37 \cdot 0,63 = 0,23$

$\sigma_f = \sqrt{0,23} = 0,48$

Notăm cu f_k v.a. care mon. ev. A la pasul k. $f_1, \dots, f_{500}$ indep. și distribuite ca f

c) $P(300 \le f_1 + \dots + f_{500} \le 400) \overset{TLC}{=} \Phi\left(\frac{400 - 500 \cdot 0,63}{5\sqrt{10} \cdot 0,48}\right) -$

$- \Phi\left(\frac{300 - 500 \cdot 0,63}{5\sqrt{10} \cdot 0,48}\right) = \Phi\left(\frac{400 - 315}{7,55}\right) - \Phi\left(\frac{300 - 315}{7,55}\right) =$

$= \Phi\left(\frac{85}{7,55}\right) - \Phi\left(-\frac{15}{7,55}\right) = \Phi(11,2) - 1 + \Phi(1,98) = 0,98$

~~3.03.2015~~

CURS 8

~~Teorema limitei centrale~~

Forma Liapunov

(X_k) $k \ge 1$ cămp de pătr. integ.

=) n ≥ 2283

31.03.2015

acest ev.

$\Phi($ 500 -

27. 04. 2015

SEMINAR 7

1) Fie $f(\omega)$ o v.a. care mod. nr. de turiști cazați zilnic într-un hotel. $f(\omega) \in N(m, \sigma)$. Se cere $\sigma = 5$. Se face un sondaj de vol. $n = 10$: $S = (70, 68, 65, 60, 80, 75, 80, 70, 65, 80)$. Se cere un interval de încredere pt. medie cu încredere $1 - \alpha = 0,95 \iff \alpha = 0,05$

$$\bar{x} = \frac{140 + 60 + 130 + 68 + 260 + 75}{10} = \frac{570 + 183}{10} = \frac{753}{10} \Rightarrow$$

$$\bar{\sigma}^2 = \frac{1 + 9 + 36 + 121 + 81 + 16 + 81 + 1 + 36 + 81}{10} \Rightarrow$$

$\bar{\sigma}^2 = \frac{167 + 253 + 53}{10} = \frac{463}{10} = 46$

$I = \left[\bar{x} - \frac{\bar{\sigma}}{\sqrt{n}} z_{1-\frac{\alpha}{2}} ; \bar{x} + \frac{\bar{\sigma}}{\sqrt{n}} z_{1-\frac{\alpha}{2}} \right]$.

$\frac{\bar{\sigma}}{\sqrt{n}} z_{1-\frac{\alpha}{2}} = \frac{5}{\sqrt{10}} \cdot z_{0,975} = \frac{5 \cdot 1,96}{\sqrt{10}} \simeq 3,1$

$I = [68,2 ; 74,4]$

---

2) La face cântăriea unui elev cu 6 cântare. $S = (1,02; 0,99; 1,01; 0,98; 1,02; 1,03)$. Beanul de cântărire furnizează o v.a. norm. distr. $f(w) \in N(m, \sigma)$.

a) $\bar{x} = ?$, $\bar{\sigma} = ?$, $s = ?$

b) Det. un interval de încr. pt. media cu încr. $1 - \alpha = 0,95$

c) Det. un int. de încr. pt. σ cu încr. $1 - \alpha = 0,95$.

a) $\bar{x} = \frac{2,04 + 2 + 2,01}{6} = \frac{6,05}{6} \simeq 1,01$

$\bar{\sigma}^2 = \frac{2 \cdot 0,01^2 + 2 \cdot 0,02^2 + 0,03^2}{6} = 0,000317 \Rightarrow \bar{\sigma} = 0,0178$

$s^2 = 0,00038 \Rightarrow s \simeq 0,0195$

b) $I = \left[\bar{x} - \frac{s}{\sqrt{n}} t_{1-\frac{\alpha}{2}; n-1} ; \bar{x} + \frac{s}{\sqrt{n}} t_{1-\frac{\alpha}{2}; n-1} \right]$

$\frac{s}{\sqrt{n}} \cdot t_{1-\frac{\alpha}{2}, n-1} \overset{\simeq 2,57}{\simeq} 0,0205$

$I = [0,9895; 1,0305]$

c) $I = \left[\frac{\sqrt{n-1} \cdot s}{\chi_{1-\frac{\alpha}{2}, n-1}} ; \frac{\sqrt{n} \cdot s}{\chi_{\frac{\alpha}{2}; n-1}} \right]$

$\frac{s}{\chi_{1-\frac{\alpha}{2}; n-1}} = \frac{0,0195}{12,83} = 0,00152$; $\frac{s}{\chi_{\frac{\alpha}{2}; n-1}} = 0,0235$

$I = [3,5 \cdot 10^{-3} ; 5,76 \cdot 10^{-2}]$

Gf. TR, $I = \frac{2}{i} \cdot 2\pi i \cdot \operatorname{Rez} f(-\frac{1}{3}) = 4\pi \cdot \lim_{z \to -\frac{1}{3}} \frac{z+\frac{1}{3}}{3z^2+10z+3} \overset{L'H}{=}$

$= 4\pi \lim_{z \to -\frac{1}{3}} \frac{1}{6z+10} = 4\pi \cdot \frac{1}{6(-\frac{1}{3})+10} = \frac{\pi}{2}$

12.05.2016

SEMINAR 9

1) $\int_0^{2\pi} \frac{1}{3+\cos x} dx$

$z = e^{ix}$; $\cos x = \frac{e^{ix}+e^{-ix}}{2} = \frac{z+\frac{1}{z}}{2} = \frac{z^2+1}{2z}$

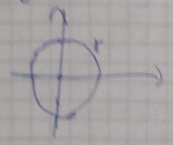
$dz = ie^{ix}dx = iz\,dx$

$\int_0^{2\pi} \frac{dx}{3+\cos x} = \int_\gamma \frac{1}{3+\frac{z^2+1}{2z}} \cdot \frac{dz}{iz} =$

$= \int_\gamma \frac{2z}{z^2+6z+1} \cdot \frac{dz}{iz} = -2i \int_\gamma \frac{dz}{z^2+6z+1} = -i \cdot 4\pi i \cdot \operatorname{Rez}(f, -3+2\sqrt{2}) =$

$\Delta = 36-4 = 32$

$z_{1,2} = \frac{-6 \pm 4\sqrt{2}}{2}$; $z_1 = -3+2\sqrt{2}$, $z_2 = -3-2\sqrt{2}$



$= 4\pi \cdot \lim_{z \to -3+2\sqrt{2}} \frac{z+3-2\sqrt{2}}{z^2+6z+1} = 4\pi \cdot \lim_{z \to -3+2\sqrt{2}} \frac{1}{2z+6} =$

$= 2\pi \cdot \frac{1}{2\sqrt{2}} = \frac{\pi\sqrt{2}}{2}$

$\operatorname{tg}\frac{x}{2} = \frac{\sin\frac{x}{2}}{\cos\frac{x}{2}} = \frac{\sqrt{\frac{1-\cos x}{2}}}{\sqrt{\frac{1+\cos x}{2}}} = \sqrt{\frac{1-\cos x}{1+\cos x}} \Rightarrow$

$\Rightarrow \frac{1-\cos x}{1+\cos x} = \operatorname{tg}^2\frac{x}{2} \Rightarrow 1-\cos x = a(1+\cos x) \Rightarrow$

$\Rightarrow (a+1)\cos x = 1-a \Rightarrow \boxed{\cos x = \frac{1-\operatorname{tg}^2\frac{x}{2}}{1+\operatorname{tg}^2\frac{x}{2}}}$

$\int_0^{2\pi} \frac{1}{3+\frac{1-\operatorname{tg}^2\frac{x}{2}}{1+\operatorname{tg}^2\frac{x}{2}}} dx = \int_0^{2\pi} \frac{1+\operatorname{tg}^2\frac{x}{2}}{4+2\operatorname{tg}^2\frac{x}{2}} dx$

Fie $t = \operatorname{tg}\frac{x}{2} \Rightarrow dt = \frac{1}{2} \cdot \frac{1}{\cos^2\frac{x}{2}} dx \Rightarrow$

$\Rightarrow dt = \frac{1}{2} \cdot \frac{\sin^2\frac{x}{2}+\cos^2\frac{x}{2}}{\cos^2\frac{x}{2}} dx \Rightarrow$

$\Rightarrow dt = \frac{1}{2}(\operatorname{tg}^2\frac{x}{2}+1)dx = \frac{1}{2}(t^2+1)dx$

$\Rightarrow I = \int_0^{2\pi} \frac{1}{2+2\cos^2\frac{x}{2}} dx$ $\quad \cos(2\pi-x) = \cos x$

$I = 2\int_0^{\pi} \frac{1+\operatorname{tg}^2\frac{x}{2}}{2+\operatorname{tg}^2\frac{x}{2}} dx = \int_0^{\infty} \frac{1+t^2}{2+t^2} \cdot \frac{2}{1+t^2} dt =$

$= \int_0^{\infty} \frac{2}{t^2+2} dt = 2 \cdot \frac{1}{\sqrt{2}} \operatorname{arctg}\frac{t}{\sqrt{2}} \Big|_0^{\infty} = \sqrt{2} \cdot \frac{\pi}{2} = \frac{\pi\sqrt{2}}{2}$

15.05.2016

CURS 10

Cauchy (teorema reziduurilor)

$\int_\gamma f(z)dz = 2\pi i \sum_{\substack{k \text{ a.i.} \\ \text{din } \gamma}} \operatorname{Rez}(f, z_k)$

Caz particular (nu avem pct. sg. pe γ): $\int_\gamma f(z)dz = 0$ (th. lui Cauchy)

1) $\int_0^{2\pi} R(\sin t, \cos t)dt$; $R(u,v) = \frac{P(u,v)}{Q(u,v)}$

$Q(u,v) \neq 0$ pt. $u^2+v^2 = 1$

Ex.: $\int_0^{2\pi} \frac{dt}{13+12\sin t}$

$e^{it} = z$; $t \in (0, 2\pi)$; $dz = ie^{it}dt = iz\,dt$

$z = z(t) = \cos t + i\sin t$

$\sin t = \frac{e^{it}-e^{-it}}{2i} = \frac{z-\frac{1}{z}}{2i} = \frac{z^2-1}{2iz}$

$\boxed{\cos z = \frac{e^{iz}+e^{-iz}}{2}}$

Teorema reziduurilor fundamentală

Fie $f(t) \in \mathcal{O}$; $F(z) = \mathcal{L}(f(t))(z) = \frac{P(z)}{Q(z)}$

$$g(t) = \sum_{\substack{z_k \text{ pol.} \\ \text{p.f.}}} \operatorname{Rez} F(z_k)\, e^{z_k t}$$

Atunci, $f(t) = g(t)$ cu excepția a cel mult un nr. finit de pct.

Consecință: Se cunoaște $F(z) = \frac{2z+3}{z^5+z}$; $f(t) = ?$

Probleme cu transf. Laplace

1) Se dă $f(t)$, se cere $F(t)$

2) Se dă $F(z)$, se cere $f(t)$

$$f(t) = \sum_{\substack{z_k \text{ pol.} \\ \text{p.f.}}} \operatorname{Rez} F(z) \cdot e^{z_k t}$$

$z^5 + z = 0 \Rightarrow z = \begin{cases} \frac{\sqrt{2}}{2}(\pm 1 \pm i) \\ \frac{\sqrt{2}}{2}(\pm 1 \pm i) \end{cases}$

...

Calcul operațional (rez. de ec. dif. p.c. cu transf. Laplace)

Ex.: $x'' + 5x' = e^{2t} + 1$ / $\mathcal{L}$; $\begin{cases} x(0) = 0 \\ x'(0) = 1 \end{cases}$

$x''(t)\,u(t) + 5x'u(t) = e^{2t}u(t) + u(t)$

Fie $X(z) = \mathcal{L}(x(t))(z)$

$$\mathcal{L}(x''u(t))(z) + 5\,\mathcal{L}(x'u(t))(z) = \underbrace{\mathcal{L}(e^{2t}u(t))(z)}_{\frac{1}{z-2}} + \underbrace{\mathcal{L}(u(t))(z)}_{\frac{1}{z}}$$

$\mathcal{L}(x'u(t))(z) = z \cdot X(z)$

$\mathcal{L}(x''(t)u(t))(z) = z^2 \cdot X(z) - 1$

$z^2 \cdot X(z) - 1 + 5z\,X(z) = \frac{1}{z-2} + \frac{1}{z} \Rightarrow$

$\Rightarrow X(z)(z^2+5z) = \frac{1}{z-2} + \frac{1}{z} + 1 \Rightarrow$

$\Rightarrow X(z) = \frac{z^2-2}{z^2(z-2)(z+5)}$

$x(t) = \sum \operatorname{Rez} X(z) \cdot e^{zt}$; 0 pol. de ord. 2, 2 pol. de ord. 1, -5 pol. de ord. 1

$\begin{cases} x' = x + y + 1 \\ y' = x - y + t \end{cases}$ $\begin{cases} x(0) = 0 \\ y(0) = 1 \end{cases}$; $x(t) = ?$, $y(t) = ?$

$\begin{cases} x'u(t) = x\,u(t) + y\,u(t) + u(t) \\ y'u(t) = x\,u(t) - y\,u(t) + t\,u(t) \end{cases}$

$\begin{cases} z\,X(z) = X(z) + Y(z) + \frac{1}{z} \\ z\,Y(z) - 1 = X(z) - Y(z) + \frac{1}{z^2} \end{cases} \Leftrightarrow \begin{cases} (z-1)X - Y = \frac{1}{z} \\ -X + (z+1)Y = \frac{1}{z^2} + 1 \end{cases}$

$$X(z) = \frac{\begin{vmatrix} \frac{1}{z} & -1 \\ 1 + \frac{1}{z^2} & z+1 \end{vmatrix}}{\begin{vmatrix} z-1 & -1 \\ -1 & z+1 \end{vmatrix}} = \dots$$

$x(t) = \sum \operatorname{Rez} X(z) e^{zt}$

19.05.2025

SEMINAR 10

$$I = \int_0^{2\pi} \frac{\cos 2t}{\cos t + 3}\,dt; \quad z = e^{it}$$

$\cos 2t = \frac{e^{2it} + e^{-2it}}{2} = \frac{z^2 + \frac{1}{z^2}}{2} = \frac{z^4+1}{2z^2}$

$\cos t = \frac{e^{it} + e^{-it}}{2} = \frac{z + \frac{1}{z}}{2} = \frac{z^2+1}{2z}$

$$I = \int_{|z|=1} \frac{z^4+1}{2z^2} \cdot \frac{2z}{z^2+1+6z} \cdot \frac{dz}{iz} \quad \left(dz = ie^{it}dt \Rightarrow dt = \frac{dz}{iz} \right)$$

$$I = \frac{1}{i} \int_{|z|=1} \frac{z^4+1}{z^4+6z^3+z^2}\,dz \overset{\text{R. 1/2 t.}}{=} \frac{1}{i} \cdot 2\pi i \sum \operatorname{Rez} \frac{z^4+1}{z^2(z^2+6z+1)}$$

$z^4 + 6z^3 + z^2 = z^2(z^2+6z+1) =$

$= z^2(z+3-2\sqrt{2})(z+3+2\sqrt{2}) \Rightarrow \begin{cases} z_1 = 0 & \text{ord } 2 \\ z_2 = -3+2\sqrt{2} & \text{ord } 1 \\ z_3 = -3-2\sqrt{2} & \text{ord } 1 \end{cases}$

$I = 2\pi i \left[\lim_{z\to 0} \left(\frac{z^3+1}{z^2+6z+1} \right)' + \lim_{z\to -3+2\sqrt{2}} \frac{z^3+1}{z^2(z+3+2\sqrt{2})} + \right.$

$\left. + \lim_{z\to -3-2\sqrt{2}} \frac{z^3+1}{z^2(z+3-2\sqrt{2})} \right] = 2\pi i \left[\lim_{z\to 0} \frac{4z^3(z^2+6z+1) - (2z+6)(z^3+1)}{(z^2+6z+1)^2} + \right.$

$\left. + \frac{(-3+2\sqrt{2})^3+1}{(-3+2\sqrt{2})^2 \cdot 4\sqrt{2}} + \frac{(-3-2\sqrt{2})^3+1}{(-3-2\sqrt{2})^2 \cdot (-4\sqrt{2})} \right] = 2\pi i \left[\frac{-6}{1} + \frac{(17-12\sqrt{2})^2+1}{(17-12\sqrt{2})\cdot 4\sqrt{2}} + \right.$

$\left. + \frac{(17+12\sqrt{2})^2+1}{(17+12\sqrt{2})\cdot 4\sqrt{2}} \right] = 2\pi i \left(-6 + \frac{289+288+1-408\sqrt{2}}{-96+68\sqrt{2}} - \right.$

$\left. - \frac{289+288+1+408\sqrt{2}}{96+68\sqrt{2}} \right) = \dots =$

$I = \int_0^{2\pi} \frac{\cos 7t}{\cos t + 3}\, dt; \quad \cos 7t = \frac{e^{7it} + e^{-7it}}{2}$

$I = \int_0^{2\pi} \frac{\cos 7t}{\cos t + 3}\, dt \qquad J + iJ = \int_0^{2\pi} \frac{e^{7it}}{\cos t + 3}\, dt$

$J = \int_0^{2\pi} \frac{\sin 7t}{\cos t + 3}\, dt$

Se calc. $I + iJ$ și cum $I, J \in \mathbb{R} \Rightarrow I = \mathrm{Re}(I + iJ)$

21.05.20

CURS 11

Transformata Fourier discretă. Transformata Z

$\mathcal{S}$ = mulțimea semnalelor discrete

$x \in \mathcal{S}$, $x: \mathbb{Z} \to \mathbb{C} \Leftrightarrow (x_n)_{n\in\mathbb{Z}}$

$\mathcal{S}_+$ = mult. semnalelor discrete fără trecut

$x \in \mathcal{S}_+ \Leftrightarrow \begin{cases} x \in \mathcal{S} \\ x(n) = \begin{cases} 0; & n<0 \\ x_n \in \mathbb{C}; & n \geq 0 \end{cases} \end{cases}$

$\begin{cases} x, y \in \mathcal{S}; & x+y \in \mathcal{S} \\ x, y \in \mathcal{S}_+; & x+y \in \mathcal{S}_+ \\ \lambda x \end{cases}$

Exemple 1) $u \in \mathcal{S}_+$; $u_n = \begin{cases} 0; & n<0 \\ 1; & n \geq 0 \end{cases}$

2) $\delta_k \in \mathcal{S}$ $\quad \delta_k(n) = \begin{cases} 1; & n=k \\ 0; & n \neq k \end{cases}$

$\delta_k(n) \in \mathcal{S}_+ \Leftrightarrow k \geq 0$

↳ semnalul lui Dirac

$\delta_2(n) = \begin{cases} 0; & n \neq 2 \\ 1; & n = 2 \end{cases}$

Produsul de convoluție a două semnale discrete

Fie $x, y \in \mathcal{S}$; $(x * y)(n) \overset{\text{def}}{=} \sum_{i+j=n} x(i) y(j) = \sum_{\substack{i,j\geq 0 \\ i+j=n}} x_i y_j$

Obs: 1) $x * y$ nu este pusă tot definită

2) $x, y \in \mathcal{S}_+ \Rightarrow x * y \in \mathcal{S}_+$

Dem: $(x*y)(n) = \sum_{i+j=n} x_i y_j = \begin{cases} 0; & n<0 \\ x_0 y_n + \dots + x_n y_0; & n \geq 0 \end{cases}$

Ex.: $(x*y)(1) = x_0 y_1 + x_1 y_0$

De dem: 1) „$*$" nu are divizori ai lui 0 $\Leftrightarrow$ Dacă $x * y = 0 \Rightarrow x = 0$ sau $y = 0$

Fie $x, y \in \mathcal{S}_+$ cu $x * y = 0 \Rightarrow \begin{cases} x_0 y_0 = 0 \\ x_1 y_0 + x_0 y_1 = 0 \Rightarrow \\ \vdots \end{cases}$ ... aplicăm în $\mathbb{R}$

2) „$*$" are div. ai lui 0 pe $\mathbb{R}[X]$

$z^2-4z+5=0$

$\Delta = 16-20 = -4 \Rightarrow z_{1,2} = \frac{4\pm 2i}{2} \begin{cases} z_1 = 2+i \\ z_2 = 2-i \end{cases}$

$x_n = \sum \operatorname{Res} X_{(z)} \cdot z^{n-1}_{(z_k)}$

$\operatorname{Res} X_{(1)} = \lim_{z\to 1} \frac{(z^2-4z+5)z^{n-1}}{z^2-4z+5} = \frac{1}{2}$

$\operatorname{Res} X_{(2+i)} = \lim_{z\to 2+i} \frac{z^{n-1}(z^2-4z+5)(z-2-i)}{(z-1)(z^2-4z+5)} = \lim_{z\to 2+i} \frac{z^{n-1}(z^2-4z+5)}{2z-5} =$

$= \frac{(2+i)^{n-1}[(2+i)^2-4(2+i)+5]}{4+2i-5} \frac{(2+i)^{n-1}[3+4i-8-4i+5]}{2i} = \frac{(2+i)^{n-1}\cdot 0}{2i} = 0$

$\operatorname{Res} X_{(2-i)} z^{n-1} = \lim_{z\to 2-i} \frac{(z^2-4z+5)z^{n-1}(z-2+i)}{(z-1)(z^2-4z+5)} =$

$= \lim_{z\to 2-i} \frac{(z^2-4z+5)\cdot z^{n-1}}{2z^2-4} = \frac{[(2-i)^2-4(2-i)+5](2-i)^{n-1}}{2\cdot(2-i)^2-4} =$

$= \frac{[3-4i-8+4i+5](2-i)^{n-1}}{-2i} = \frac{(2-i)^{n-1}}{2i}$

$\boxed{x_n = \frac{(2-i)^{n-1}}{2i} - \frac{(2+i)^{n-1}}{2i} + \frac{1}{2}}$

X.

Metoda 2 $x_n = 4x_{n-1} - 5x_{n-2} + 1$

$x_1 = x_2 = 1$

Recurență omogenă: $\lambda^2 - 4\lambda + 5 = 0$

$\lambda_{1,2} = 2 \pm i$

Sol. generală a recurenței omogene

$x_0 = c_1 \lambda_1^n + c_2 \lambda_2^n = c_1(2+i)^n + c_2(2-i)^n$

Le aflu o sol. particulară a ec. neomogene

1) Încerc un șir constant ($x_n = k$ dacă nu ar fi gr. 1, $x_n = an+b$ ...)

$k = 4k - 5k + 1 \Rightarrow k = \frac{1}{2}$

$x_n = x_{n_0} + x_n^p = c_1(2+i)^n + c_2(2-i)^n + \frac{1}{2}$

$\begin{cases} x_1 = 1 \\ x_2 = 1 \end{cases} \Rightarrow \begin{cases} c_1(2+i) + c_2(2-i) + \frac{1}{2} = 1 \\ c_1(2+i)^2 + c_2(2-i)^2 + \frac{1}{2} = 1 \end{cases} \Rightarrow$

$\Rightarrow \begin{cases} c_1(2+i) + c_2(2-i) = \frac{1}{2} \\ c_1(3+2i) + c_2(3-2i) = \frac{1}{2} \end{cases}$

$\Delta_S = \begin{vmatrix} 2+i & 2-i \\ 3+2i & 3-2i \end{vmatrix} = 8-i-(6-3i+4i+2) = 8-i-8-i+2i = 0$

$\Delta_{S_1} = \begin{vmatrix} \frac{1}{2} & 2-i \\ \frac{1}{2} & 3-2i \end{vmatrix} = \frac{1}{2}(3-2i-2+i) = \frac{1}{2}(1-i) = 0$

$\Rightarrow x_1 = \frac{1}{2} \cdot \frac{1-i}{-2i} = \frac{-1+i}{4i} = \frac{1}{4} + \frac{1}{4}i$

$\Delta_{c_2} = \begin{vmatrix} 2+i & \frac{1}{2} \\ 3+2i & \frac{1}{2} \end{vmatrix} = \frac{1}{2}(2+i-3-2i) = \frac{1}{2}(-1-i) \Rightarrow$

$\Rightarrow x_2 = \frac{1}{4} \cdot \frac{1+i}{i} = \frac{1}{4} - \frac{1}{4}i$

$x_n = \left(\frac{1}{4} + \frac{1}{4}i\right)(2+i)^n + \left(\frac{1}{4} - \frac{1}{4}i\right)(2-i)^n + \frac{1}{2}$

26.05.2015

SEMINAR 11

1) $x_n = 4x_{n-1} + 1$; $n \geq 2$; $x_1 = 2$

a) $x_n = ?$

b) x_{100} e nr. prim

2) $x' = 4x + e^{2t}$; $x_{(0)} = 0$

$x_{(t)} = ?$

3) $F(z) = \frac{2z+1}{(z-1)^2(z+2)}$

$\operatorname{Res} F_{(z_k)} = ?$

$\Sigma \begin{cases} x' = x + y + t^2 \\ y' = x + y + t \end{cases}$ $\begin{cases} x(0) = 0 \\ y(0) = 1 \end{cases}$ $x(t) = ?$ $y(t) = ?$

---

a) $x_n = \frac{1}{2} x_{n-1} + 1$; $n \geq 2$; $x_1 = 2$

a) $x_n = ?$

Soluția $x_n = 0$; $n \leq 0$

$x_n - \frac{1}{2} x_{n-1} = 1$; $n \geq 2$

$x_n - \frac{1}{2} x_{n-1} = \begin{cases} 1; n \geq 2 \\ 2; n = 1 \\ 0; n \leq 0 \end{cases} = a_n$

$x_{(n)} - \frac{1}{2} (x * e_0)_{(n)} = a_n$

$X_{(z)} - \frac{1}{2} X_{(z)} = \mathcal{Z}_{(x_n * e_0)}(z)$

$\mathcal{Z}_{(x_n * e_0)}(z) = \sum_{n=0}^{\infty} a_n z^{-n} = \frac{2}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots = \frac{1}{z} + \frac{\frac{1}{z}}{1 - \frac{1}{z}} =$

$= \frac{1}{z} + \frac{1}{z-1} = \frac{2z-1}{z(z-1)}$

$X - \frac{1}{2} X = \frac{2z-1}{z(z-1)} \quad / \cdot z$

$(z - \frac{1}{2}) X = \frac{2z-1}{z-1} \Rightarrow X = \frac{2z-1}{(z-\frac{1}{2})(z-1)}$

$x_{(n)} = \sum \lim_{z \to z_k} X(z) z^{n-1} |_{z_k} =$

$= \sum \lim_{z \to z_k} \frac{z^{n-1}(2z-1)}{(z-\frac{1}{2})(z-1)} (z_k) = \lim_{z \to 1} \frac{z^{n-1}(2z-1)}{z-\frac{1}{2}} + \lim_{z \to \frac{1}{2}} \frac{z^{n-1}(2z-1)}{z-1} =$

$= -\frac{1}{\frac{1}{2}} + \frac{1^{n-1} \cdot 2}{\frac{1}{2}} = \frac{2 \cdot 2^{n-1} - 1}{\frac{1}{2}}$

b) $x_{(n)} = \frac{2 \cdot 2^{n} - 1}{\frac{1}{2}}$

Șirul în prod lim $\frac{1}{2}$ nu = infinitele de ori prod?

---

a) $x' = 4x + e^{2t}$; $x_{(0)} = 0$

Notăm $\mathcal{Z}_{(x_n)} = \mathcal{Z}_{(x_n)}(z)$

$z \cdot x_{(z)} = 4 z \cdot x_{(z)} + z^2 \cdot x_{(z)} \quad / \cdot \frac{1}{z}$

$z X_{(z)} - 0 = 4 X_{(z)} + \frac{1}{z-2}$

$(z-4) X = \frac{1}{z-2} \Rightarrow X = \frac{1}{(z-2)(z-4)}$

$x_{(t)} = \sum \lim_{z \to z_k} X_{(z)} \cdot e^{zt} |_{z_k} = -\frac{1}{2} e^{2t} + \frac{1}{2} e^{4t}$